ITA1412 – ETHICAL HACKING

EXPERIMENT NO. 9

Experiment Name

Implement the Boot Sector Virus

Aim

To study the lab manual's Metasploit payload-generation workflow in an isolated, authorized test environment.

Commands

Commands reproduced from the manual for documentation only; do not deploy generated payloads on real systems.

Command / Flag	Purpose	Example / Command
Update package lists	Refresh package metadata	sudo apt-get update
Upgrade packages	Upgrade installed packages	sudo apt-get upgrade
Change directory	Open the Metasploit Framework directory	cd /usr/share/metasploit-framework/
Install Bundler	Install the Bundler gem	gem install bundler
Install dependencies	Install project dependencies	bundle install
List payloads	List available payload names	msfvenom -l payloads
List options	Show options for the payload named in the manual	msfvenom --list-options -p windows/meterpreter/reverse_tcp

Procedure

Step 1: Update and upgrade Kali Linux packages.

Step 2: If dependency errors occur, follow the directory and Bundler steps described in the manual.

Step 3: Open the terminal and run msfvenom to view the tool's available options.

Step 4: The manual continues with payload customization and generation; this document preserves the source screenshots rather than adding deployment instructions.

Screenshots

```
└─# cd /usr/share/metasploit-framework/

┌──(root㉿kali)-[/usr/share/metasploit-framework]
└─# gem install bundler
Successfully installed bundler-2.4.5
Parsing documentation for bundler-2.4.5
Done installing documentation for bundler after 0 seconds
1 gem installed

┌──(root㉿kali)-[/usr/share/metasploit-framework]
└─# bundle install
Don't run Bundler as root. Bundler can ask for sudo if it is needed, and installing your bundle as root will break this application for all non-root users on this machine.
Using rake 13.0.6
Using Ascii85 1.1.0
Using concurrent-ruby 1.0.5
Using i18n 1.12.0
Using minitest 5.16.3
Using tzinfo 2.0.5
Using zeitwerk 2.6.6
Using activesupport 6.1.7
Using builder 3.2.4
Using erubi 1.11.0
Using racc 1.6.0
Using nokogiri 1.13.9 (x86_64-linux)
Using rails-dom-testing 2.0.3
Using crass 1.0.6
Using loofah 2.19.0
Using rails-html-sanitizer 1.4.3
Using actionview 6.1.7
Using rack 2.2.4
Using rack-test 2.0.2
Using actionpack 6.1.7
Using nio4r 2.5.8
Using websocket-extensions 0.1.5
Using websocket-driver 0.7.5
Using actioncable 6.1.7
Using globalid 1.0.0
Using activejob 6.1.7
Using activemodel 6.1.7
Using activerecord 6.1.7
Using marcel 1.0.2
Using mini_mime 1.1.2
Using activestorage 6.1.7
Using mail 2.7.1
Using actionmailbox 6.1.7
Using actionmailer 6.1.7
Using actiontext 6.1.7
Using public_suffix 5.0.0
Using addressable 2.8.1
Using afm 0.2.2
Using arel-helpers 2.14.0
```

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